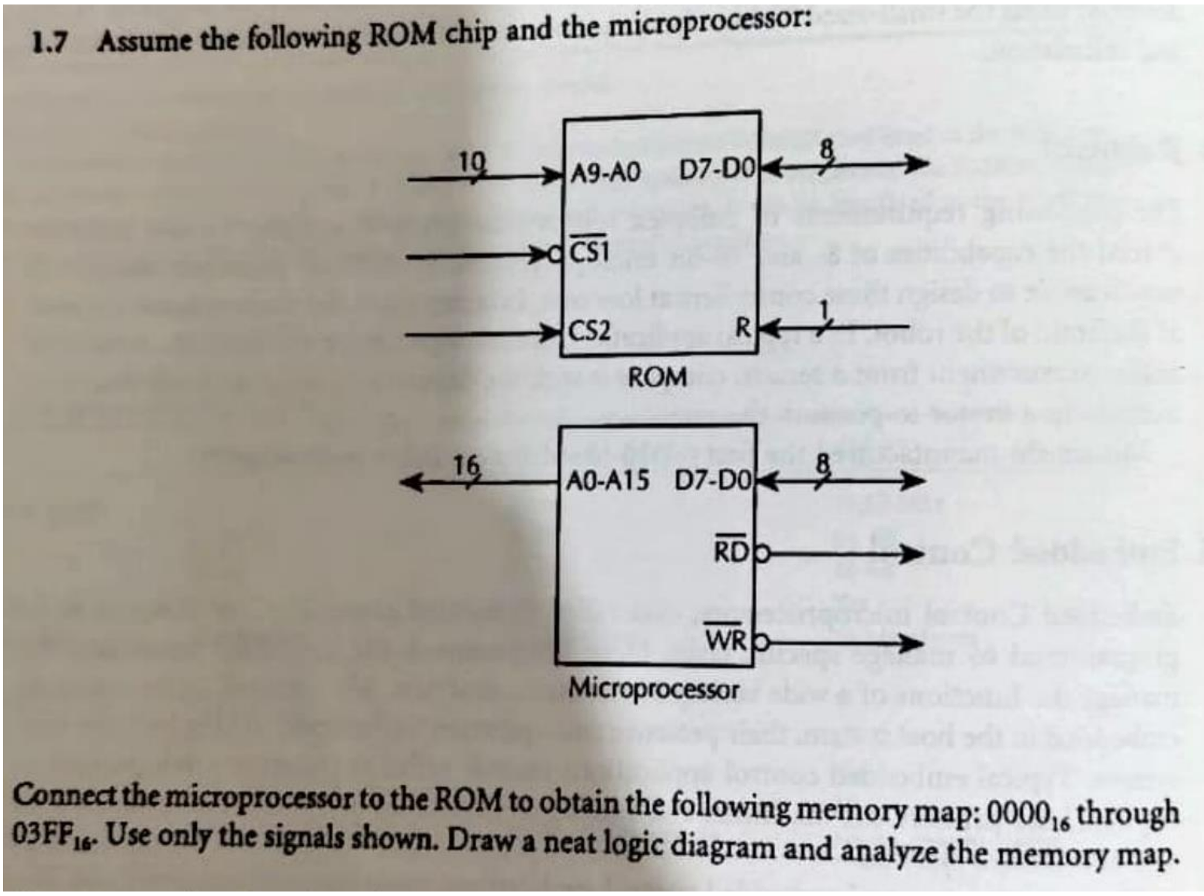


1.7 Assume the following ROM chip and the microprocessor:

Connect the microprocessor to the ROM to obtain the following memory map: 0000_{16} through $03FF_{16}$. Use only the signals shown. Draw a neat logic diagram and analyze the memory map.

To connect the microprocessor to the ROM and achieve the memory map **0000_{16}** to **$03FF_{16}$**, follow these steps:

Logic Diagram

```
Microprocessor (µP)                      ROM
-----------                             -----
A0-A9     ---------------------------> A0-A9
D7-D0     <--------------------------> D7-D0
RD        ---------------------------> R
A10-A15 --> [6-input NOR] --> CS1
            [NOR Output] ----> [NOT] --> CS2
```

Analysis

1. **Address Lines**:
   - **A0-A9** (10 lines) connect directly to the ROM. These address the 1KB (1024 bytes) of ROM space.

- **A10-A15** (6 lines) are used for chip selection. These must be **0** to select the ROM (address range **0000_{16}** to **$03FF_{16}$**).

2. **Data Lines**:
   - **D7-D0** (8-bit bidirectional bus) connects directly between the microprocessor and ROM.
3. **Control Signals**:
   - **RD** (Read) from the microprocessor connects to **R** (Output Enable) of the ROM.
   - **WR** is unused (ROM is read-only).
4. **Chip Select Logic**:
   - A **6-input NOR gate** decodes **A10-A15**:
     - Outputs **1** (high) only when **A10-A15 = 000000_2**.
   - **CS1** (active high) connects directly to the NOR gate output.
   - **CS2** (active low) connects to the NOR gate output via an **inverter**.

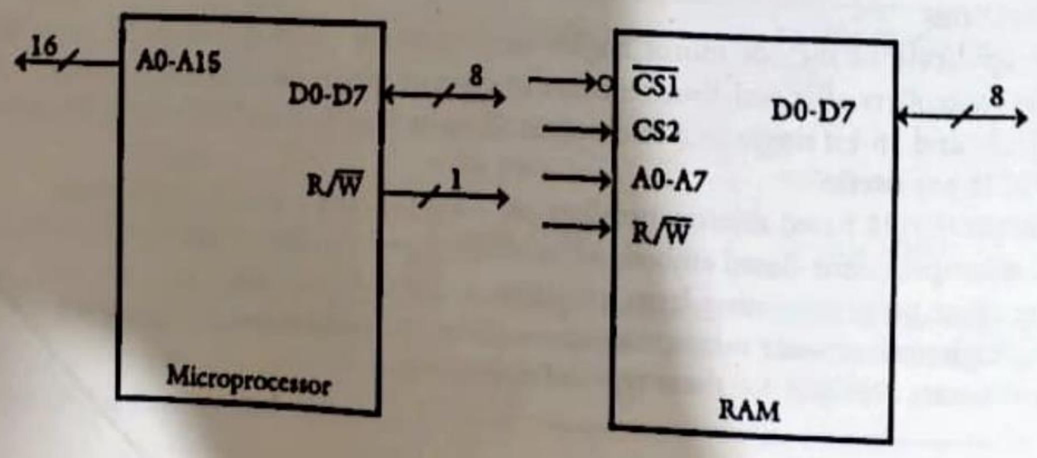
Memory Map Explanation

- **Address Range**: **0000_{16}** to **$03FF_{16}$** (1KB).
- **Decoding**:
  - **A10-A15 = 0**: Activates the NOR gate, enabling the ROM.
  - **A0-A9**: Selects the specific byte within the ROM.
- Any address outside this range (**A10-A15 $\neq$ 0**) deactivates the ROM.

Key Takeaways

- The ROM is mapped to the first 1KB of memory.
- Address decoding ensures the ROM responds only to **0000_{16}** to **$03FF_{16}$**.
- No external memory or logic is required beyond the NOR gate and inverter.

1.8 Assume the following microprocessor and the RAM chip:



Draw a neat logic diagram showing connections between the above microprocessor and the RAM chip using only the signals shown to include the memory map 0200_{16} through $02FF_{16}$. Use linear decoding.

To connect the microprocessor to the RAM chip for the memory map **0200_{16}** to **$02FF_{16}$** using **linear decoding**, follow these steps:

Logic Diagram

```
Microprocessor (µP)                    RAM
-----------                           -----
A0-A7     -------------------------> A0-A7
D0-D7     <------------------------> D0-D7
R/W       -------------------------> R/W
A9        -------------------------> CS1 (active high)
A8        -------------------------> [Inverter] ---> CS2 (active low)
```

Analysis

1. **Address Lines**:
   - **A0-A7** (8 lines) connect directly to the RAM. These address the 256 bytes (00h-FFh) within the RAM.
   - **A8** and **A9** are used for chip selection:

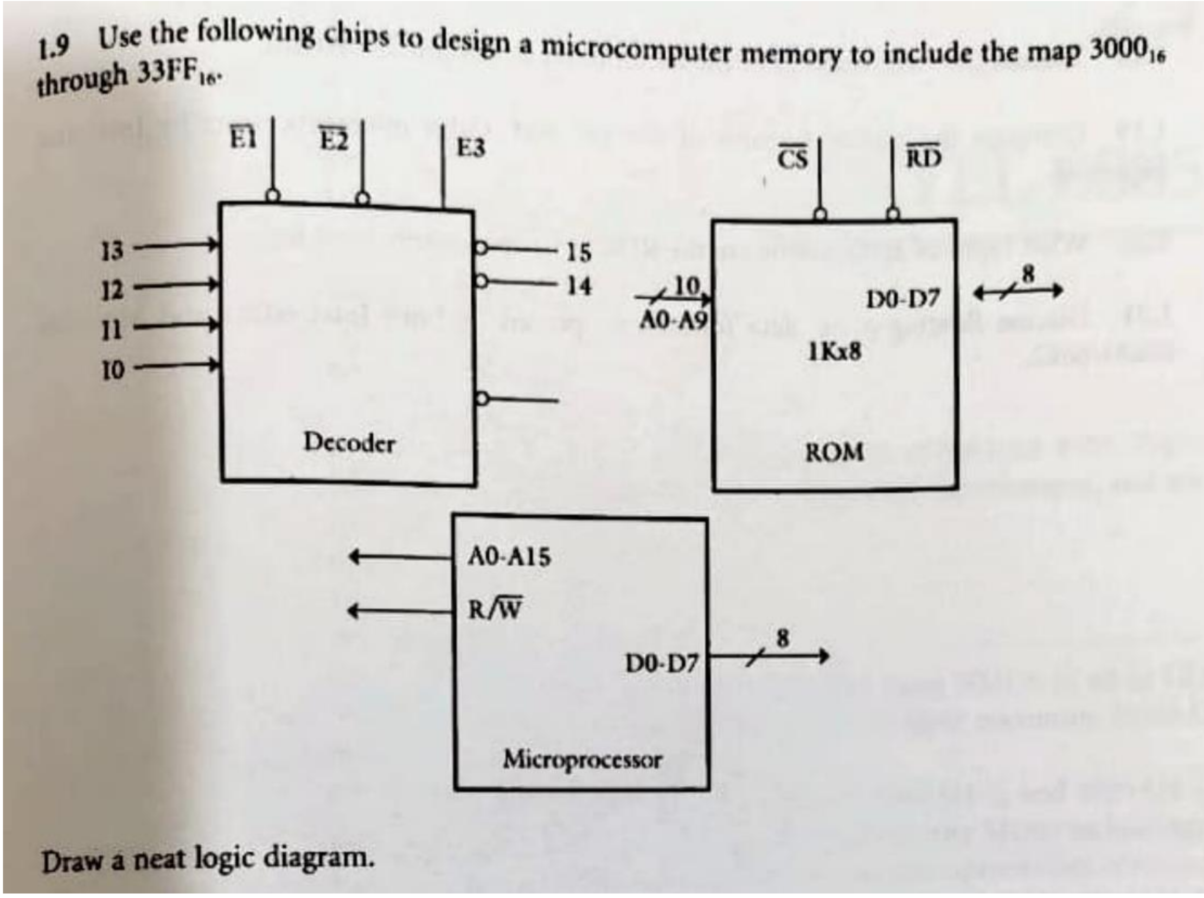
    - **A9** is connected to **CS1** (active high).
    - **A8** is inverted and connected to **CS2** (active low).
2. **Data Lines**:
    - **D0-D7** (8-bit bidirectional bus) connects directly between the microprocessor and RAM.
3. **Control Signals**:
    - **R/W** (Read/Write) connects directly to the RAM's **R/W** pin.
4. **Chip Select Logic**:
    - **CS1** is active when **A9 = 1**.
    - **CS2** is active when **A8 = 0** (via inverter).
    - The RAM is enabled only when **A9 = 1** and **A8 = 0**.

Memory Map Explanation

- **Address Range**: **0200_{16}** to **$02FF_{16}$** (256 bytes).
    - **A9 = 1** and **A8 = 0** (upper bits: **A15-A10** are ignored in linear decoding).
    - **A0-A7** select the byte within the RAM.
- **Aliasing**: The RAM will also respond to addresses like **1200_{16}**, **2200_{16}**, etc., where **A9 = 1** and **A8 = 0** (linear decoding does not check higher address bits).

Key Takeaways

- The RAM is mapped to **0200_{16}** to **$02FF_{16}$** using **A9** and **A8** for chip selection.
- Linear decoding simplifies the design but allows address aliasing.
- No additional logic (e.g., decoders) is required beyond an inverter.


1.9 Use the following chips to design a microcomputer memory to include the map 3000_{16} through $33FF_{16}$.

Draw a neat logic diagram.

To design a microcomputer memory system mapping **3000_{16} to $33FF_{16}$** using linear decoding, follow these steps:

Logic Diagram

```
Microprocessor (µP)                  RAM Chips (E1, E2, E3)
-----------                         --------------------
A0-A9     ---------------------------> A0-A9 (All RAMs)
D0-D7     <--------------------------> D0-D7 (All RAMs)
R/W       ---------------------------> R/W (All RAMs)

A15 (MSB)  }
A14        }--> [3-to-8 Decoder] --> E1, E2, E3 (Chip Selects)
A12        }
```

Analysis

1. **Address Lines**:
   - **A0-A9** connect to all RAM chips for 1KB addressing (10 bits).
   - **A15, A14, A12** connect to the decoder (pins 15, 14, 10) for chip selection.
2. **Data Lines**:
   - **D0-D7** connect bidirectionally between the microprocessor and RAMs.
3. **Control Signals**:
   - **R/W** connects to all RAMs for read/write operations.
4. **Chip Select Logic**:
   - **Decoder Inputs**:
     - **A15=0**, **A14=0**, **A12=1** (to match the range **3000_{16}**-high bits).
   - **Decoder Outputs**:
     - **E1, E2, E3** enable the RAMs when the decoder activates (e.g., **E1** for **3000_{16}-$33FF_{16}$**).

Memory Map Explanation

- **Address Range**: **3000_{16}** to **$33FF_{16}$** (1KB).
- **Linear Decoding**:
  - **A15=0**, **A14=0**, **A12=1** (ignores **A13, A11, A10** for simplicity).
  - **A0-A9** select the byte within the RAM.
- **Aliasing**: The RAMs also respond to mirrored addresses (e.g., **1000_{16}-$13FF_{16}$**) due to undecoded higher bits.

Key Takeaways

- **Decoder Configuration**: Uses **A15, A14, A12** to activate the RAMs for the desired range.
- **Data/Address Bus**: Shared across all RAMs, with chip selects managed by the decoder.
- **Linear Decoding Trade-off**: Simplifies design but allows address aliasing.

This setup ensures the RAMs are mapped to **3000_{16}-$33FF_{16}$** while adhering to the constraints of linear decoding.